

**Paper Practice 1.3 • WORKED KEY****Properties of Real Numbers**

**WORKED KEY (odd-numbered items) – Show all of your work in the space provided. When a problem asks for a property, write its FULL name. Circle your final answers.**

- 1.** Name the property of equality shown by each statement. (Properties of Equality)

**a.** If  $14 + 6 = 20$ , then  $20 = 14 + 6$ .

**Answer:** Symmetric

**b.**  $n + 9 = n + 9$

**Answer:** Reflexive

**c.** If  $x = 7$  and  $7 = 3 + 4$ , then  $x = 3 + 4$ .

**Answer:** Transitive

- 2.** Name the property shown by each statement. (Number Properties)

**a.**  $23 \cdot 1 = 23$

**b.**  $8 + (-8) = 0$

**c.**  $(2/5)(5/2) = 1$

**d.**  $17 \cdot 0 = 0$

- 3.** Evaluate:  $6 - 2^2 + 9(1)$  (Evaluate)

**Answer:**  $6 - 4 + 9 = 11$

- 4.** Evaluate:  $(3/4)(4/3) \cdot 12$  (Evaluate)

- 5.** Evaluate:  $(18 - 2 \cdot 3 + 5) \cdot 0$  (Evaluate)

**Answer:** 0 – by the Multiplicative Property of Zero you never have to simplify inside

- 6.** Use the commutative and associative properties to evaluate  $25 + 38 + 75 + 62$ . Show the regrouping. (Mental Math)

- 7.** Use the commutative and associative properties to evaluate  $4 \cdot 17 \cdot 25$ . Show the regrouping. (Mental Math)

**Answer:**  $(4 \cdot 25) \cdot 17 = 100 \cdot 17 = 1700$

- 8.** Use the properties to evaluate  $36 + 59 + 64 + 41$ . Show the regrouping. (Mental Math)

- 9.** Jade spends \$41, \$36, \$39, and \$44 on groceries over four weeks. Use one of the properties to find her total, and name the property you used. (Apply)

**Answer:**  $(41 + 39) + (36 + 44) = 80 + 80 = \$160$ .  
Commutative and Associative Properties of Addition.

- 10.** Name the property used in each step. (Justify)

$$\begin{aligned} &12 + 45 + 8 + 55 \\ &= 12 + 8 + 45 + 55 \\ &= (12 + 8) + (45 + 55) \\ &= 20 + 100 = 120 \end{aligned}$$

- 11.** Error Analysis – Gus said that  $(5 + 3) + 7 = 5 + (3 + 7)$  shows the Commutative Property. Identify his mistake. (Error Analysis)

**Answer:** The numbers stayed in the same ORDER – only the parentheses moved. That is the ASSOCIATIVE Property. Commutative changes order; associative changes grouping.

- 12.** Write your own example of the Multiplicative Inverse Property, then explain in one sentence why the product is always 1. (Explain)